NVIDIA A100 Tensor Core GPU 80GB - DATASHEET

Product Status: NRND (Not Recommended for New Designs)

Overview

The NVIDIA A100 80GB is a data center GPU based on the Ampere architecture, designed for AI training, inference, and HPC workloads. It delivers up to 20x higher performance over the prior V100 generation. Features Multi-Instance GPU (MIG) technology for up to 7 isolated GPU instances. Status: NOT RECOMMENDED FOR NEW DESIGNS - superseded by H100.

Compute Specifications

CUDA Cores: 6,912. Tensor Cores: 432 (3rd generation). Peak FP64: 19.5 TFLOPS. Peak FP32: 19.5 TFLOPS. Peak TF32 Tensor Core: 156 TFLOPS. Peak FP16 Tensor Core: 312 TFLOPS. Peak INT8 Tensor Core: 624 TOPS. Peak BF16: 312 TFLOPS. Sparsity support: 2:4 structured sparsity doubles effective throughput.

Memory and Interconnect

HBM2e memory: 80 GB. Memory bandwidth: 2,039 GB/s. Memory bus width: 5120-bit. ECC: full hardware ECC. L2 cache: 40 MB. NVLink 3.0: 12 links, 600 GB/s total bidirectional bandwidth. PCIe Gen4 x16 host interface. NVSwitch support for up to 8 GPUs fully connected.

Power and Thermal

TDP: 400W (SXM4), 300W (PCIe). Max GPU temperature: 83 degrees C. Cooling: passive heatsink (requires system-level airflow of minimum 70 CFM). Form factors: SXM4 (for DGX/HGX systems), PCIe dual-slot. Operating temperature: 0 to 50 degrees C ambient.

Migration to H100

The A100 is being phased out in favor of the H100 (Hopper architecture). The H100 provides: 80 GB HBM3 at 3,350 GB/s bandwidth (1.6x improvement), FP8 Tensor Cores at 3,958 TOPS, PCIe Gen5, and 2nd-gen MIG with up to 7 instances. The H100 is a functional drop-in replacement for A100 in most AI training frameworks.